2010376103 CSE4261 Assignment-5

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June 2025

Github code link

1 Dataset

The experiments use a single ImageNet example image. The original image was downloaded and decoded with TensorFlow utilities, resized to the model input size (224×224) and preprocessed so it matches MobileNet's expected input range and normalization. Predictions and all manipulations (FGSM perturbation and Gaussian noise) were performed on this preprocessed image tensor.

2 Experimental Setup

- **Model:** MobileNet with ImageNet weights (`include_top=True`). Model set to non-trainable for inference-only experiments.
- **Target image:** single ImageNet example (green mamba). Input shape after preprocessing: (1,224,224,3).
- **Adversarial method:** Fast Gradient Sign Method (FGSM). We compute gradient of categorical cross-entropy loss w.r.t. the input image and form perturbation as sign of gradients. Adversarial examples are created as:
$$x_{adv} = \text{clip}(x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(f(x), y_{target})), -1, 1).$$
- **Epsilons tested:** $\epsilon = [0, 0.01, 0.1, 0.15]$.
- **Gaussian noise test:** additive Gaussian noise with combinations of mean $\in \{0.0, 0.2, 0.4\}$ and stddev $\in \{0.0, 0.3, 0.6\}$. Noisy images are clipped to the model input range $[-1, 1]$.
- **Evaluation:** For each perturbed image we obtain model prediction and top-1 class confidence using MobileNet's decode utilities.
- **Visualization:** perturbation map and grids of predicted images are saved as figures ('img/grad.png', 'img/gauss.png') and included below.

3 Attack Implementation (brief)

Implementation follows standard FGSM:

1. Preprocess input image for MobileNet.
2. Create one-hot target label vector (here index = 64 for green mamba).
3. Use `tf.GradientTape()` to compute gradient of CategoricalCrossentropy w.r.t. input.
4. Form signed gradient and apply scaled perturbation for each ϵ.
5. Clip outputs to valid input range.

For Gaussian noise, we sample $\mathcal{N}(\mu, \sigma^2)$ with specified μ, σ and add to the preprocessed image followed by clipping.

4 Results & Discussion

Table 1: Classification results under FGSM attack with varying epsilon values.

Epsilon (ϵ)	**Predicted Class**	**Confidence (%)**
0.00	green_mamba	95.73
0.01	horned_viper	58.67
0.10	horned_viper	46.93
0.15	rock_python	21.36

Table 2: Classification results under Gaussian noise with varying mean and stddev.

Mean	**Stddev**	**Predicted Class**	**Confidence (%)**
0.0	0.0	green_mamba	95.73
0.0	0.3	green_snake	22.27
0.0	0.6	jean	6.04
0.2	0.0	green_mamba	95.80
0.2	0.3	boa_constrictor	10.21
0.2	0.6	velvet	28.71
0.4	0.0	green_mamba	96.44
0.4	0.3	green_snake	27.52
0.4	0.6	velvet	30.28

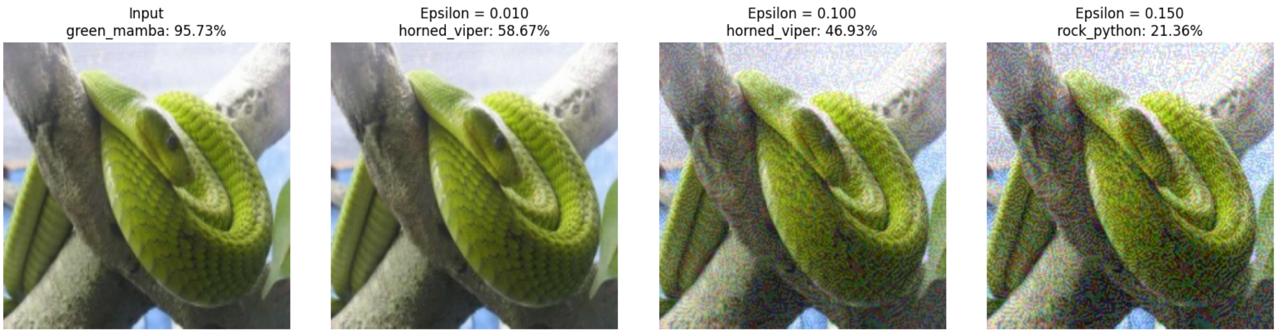

Figure 1: FGSM adversarial attack. Top row: original and adv images for different ϵ. Bottom row: perturbation (signed gradient) visualized.

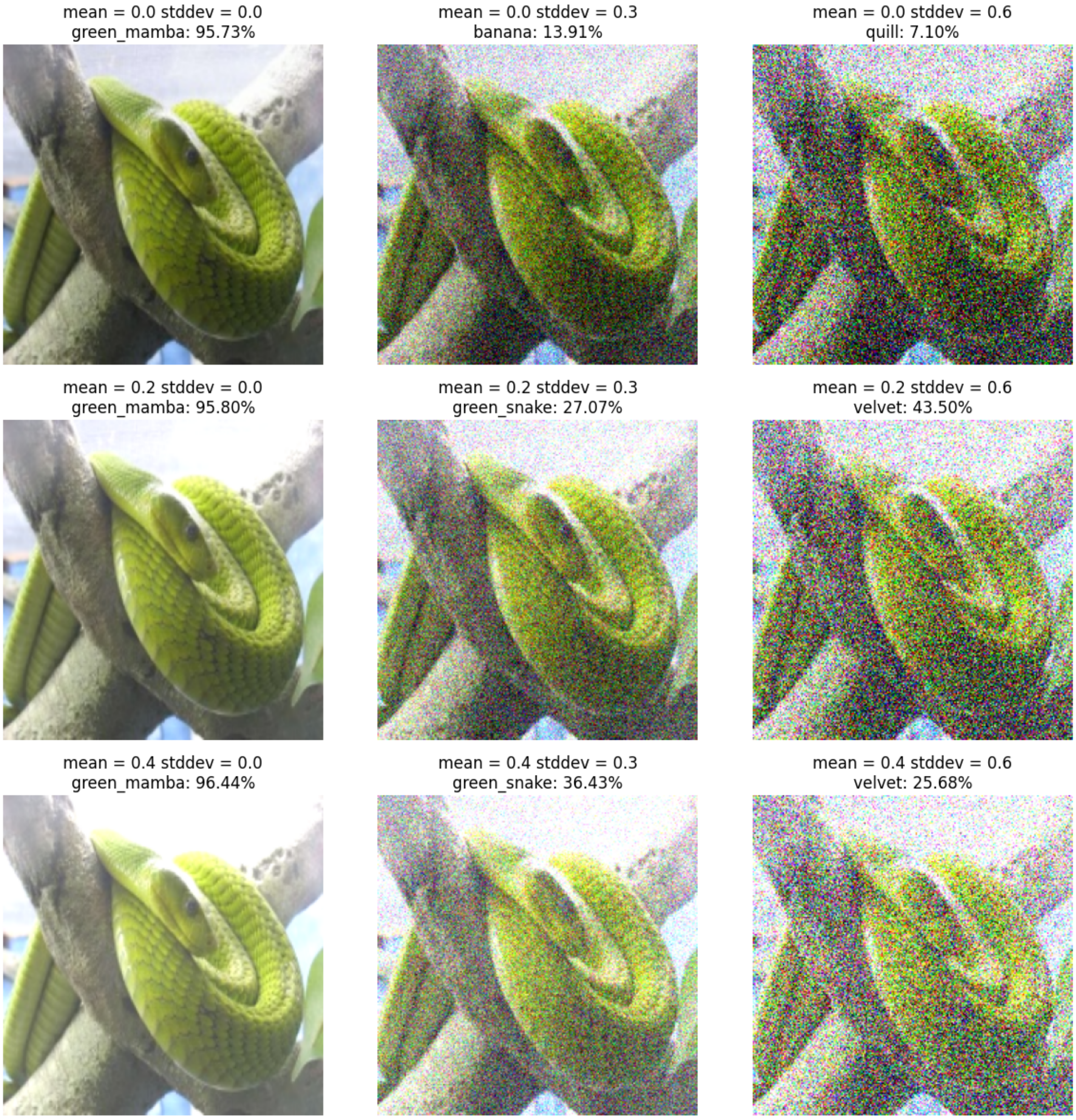

Figure 2: Gaussian noise perturbation grid. Rows vary mean, columns vary stddev.

- **Effectiveness of FGSM:** Even very small ϵ (0.01) changes the top-1 prediction from `green_mamba` (95.7% confidence) to `horned_viper` (58.7%). Larger ϵ further reduce confidence and often change class. This shows FGSM is highly effective because it uses model gradients to create targeted, structured perturbations that maximize loss.

- **Gaussian noise vs FGSM:** Random Gaussian noise can change predictions only when its magnitude (stddev) is large. Results show that to sufficiently alter prediction confidence or class, stddev must be high (e.g., 0.6), which produces visibly corrupted images. FGSM achieves misclassification with imperceptibly small perturbations. Gaussian noise is less predictable and less efficient at producing targeted misclassification.

- **Interpretation:** FGSM exploits the model's decision boundary by following the local gradient direction. Gaussian noise lacks this alignment and therefore requires larger amplitude to cross decision boundaries. Consequently FGSM is a more practical threat for stealthy adversarial attacks.

5 Conclusion

The implemented FGSM reliably produces adversarial examples that change MobileNet predictions at very small perturbation budgets. Adding Gaussian noise can also alter predictions but typically requires much larger noise magnitude and leads to visible corruption. Structured adversarial perturbations are thus substantially more effective than random Gaussian noise for evading pretrained CNNs.